Project

One-Pager

Project Title: Knowledge Graph-based APT Detection in Enterprise Networks
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> I have used AI for generating and collecting some ideas I had. I found some datasets (https://csr.lanl.gov/data/cyber1/) that seem worthy of giving effort to this topic. If, on the other hand, this turns out to be a greater take than imagined - I will default back to the fallback project.

Motivation

Advanced Persistent Threats (APTs) are sophisticated, multi-stage cyberattacks where an intruder establishes an illicit, long-term presence on a network. Modern Security Information and Event Management (SIEM) systems struggle to correlate low-level log events (e.g., a file download followed by an unusual registry change) into a high-level attack narrative.

Problem

The core challenge is "connecting the dots." An attacker might use valid credentials to move laterally through a network. To detect this, we need to model the relationships between **Users**, **Processes**, **Files**, and **Network Connections**. We need a KG that can distinguish between "Business-as-Usual" and a "Stealthy Attack Chain."

Solution

The solution is a hybrid KG architecture:

- **Logical Component:** Uses a rule-based reasoner to encode the **MITRE ATT&CK** framework (e.g., "If a process creates a remote thread in a system process, flag as Potential Code Injection").
- **KGE Component:** Uses embeddings (like **TransE** or **RotatE**) to map entities into a vector space. Anomalous behavior is detected when a "Link" (e.g., User → Access → Server) has a very low plausibility score compared to historical baseline embeddings.
  The output of the logical rules feeds into the embedding model as additional "provenance" metadata to refine the detection accuracy.

Steps

- **Data Integration:** Combine the **LANL (Los Alamos National Lab)** Cybersecurity dataset (network flows/auth events) with the **MITRE ATT&CK** structured knowledge base.
- **Logical Reasoning:** Formulate rules in a language like **Datalog** (or using the **Vadalog** system) to represent the stages of a cyber-kill chain.
- **Embeddings:** Use the **PyKEEN** or **DGL-KE** library to implement KGE method **Z** (e.g., ComplEx) to perform link prediction for "Malicious Connection" detection.

Scope

The LANL dataset is massive, so the scope will be limited to a specific subset of "Red Team" events (simulated attacks). The project will focus on the **integration** of these two components rather than inventing a new embedding algorithm.

Category	LO Reference	Focus in this Project
Primary Focus	**LO1, LO2**	Applying **KGEs** to predict malicious links and using **Logical Knowledge** to model attack patterns.
Proficiency	**LO4**	Comparing **Graph Data Models** (Property Graphs for logs vs. RDF for MITRE knowledge).
	LO5	Designing the **Architecture** that connects raw log ingestion to the Vadalog reasoning engine.
	LO6, LO8	Using **Scalable Reasoning** for real-time log analysis and **KG Evolution** to update the graph as new logs

		arrive.
	LO11, LO12	Providing **Services** (alerts/dashboards) and describing the **Connection** between symbolic logic and ML-based anomaly detection.
	LO9	Describing the **Real-World Application** of KGs in Security Operations Centers (SOCs).
Excluded	**LO3, LO10**	This project will not use **GNNs** or focus on **Financial** specific data.